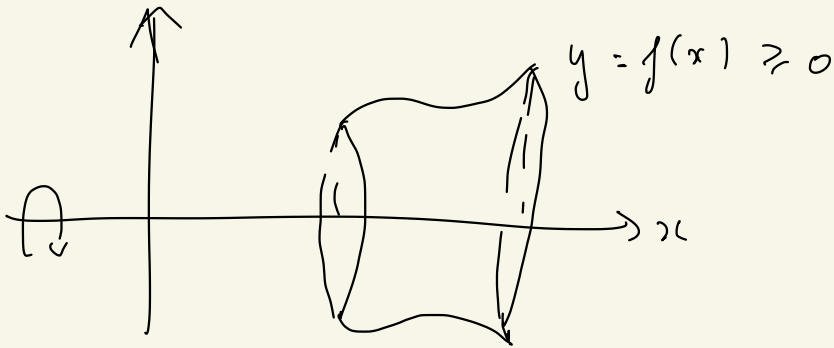
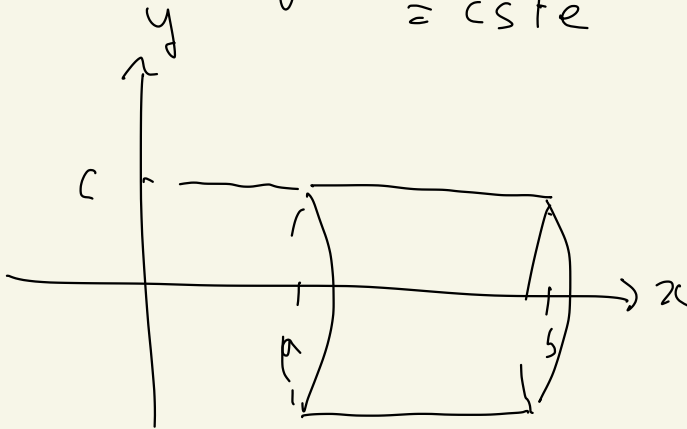


Areas of surfaces of revolution

Calculus A(1)
12/12



① Case $f(x) = c > 0$
 $= \text{cte}$

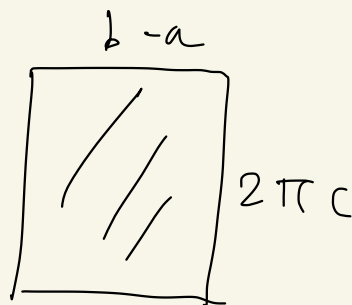


$$A = 2\pi c \cdot (b-a)$$

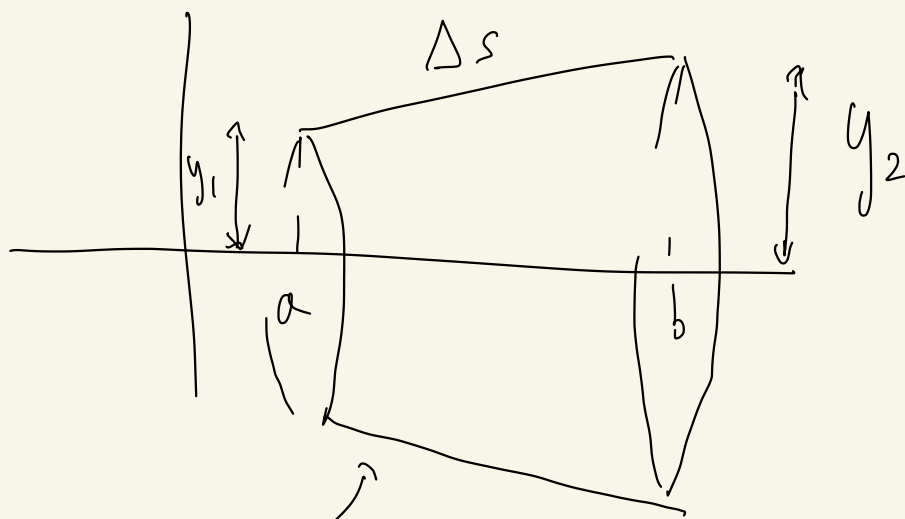
Cut and roll out

$\leadsto$ rectangle

$$A = (b-a)2\pi c$$



(2) Case $f(x) = mx + c$



part of a cone.

Classical formula:

$$A = 2\pi y^* \Delta s$$

$$y_1 = y_2$$

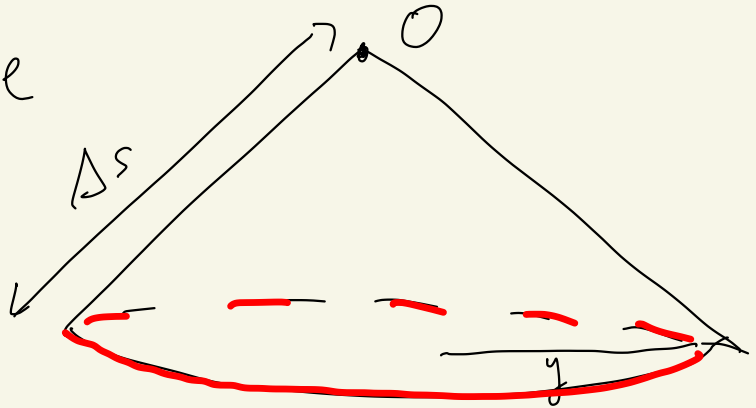
$\leadsto$

$$A = 2\pi y_1 \Delta s$$

where $y^* = \frac{y_1 + y_2}{2}$

Explanation :

Let us start with the area of a cone



y = radius of the base.

Cut and roll out :

One gets a sector of a disc as follows :



One easily gets :

$$A = \pi \cdot y \cdot \Delta s$$

So Area of our cone

$$= A = \pi y \Delta s$$

$$= \frac{2\pi y + 2\pi r_0}{2} \cdot \Delta s$$

general case for
part of a cone



A = area of the part of the

cone

$$= \pi(l + \Delta s) \cdot y_2$$

$$- \pi \cdot l \cdot y_1$$

$$\stackrel{?}{=} \pi(y_1 + y_2) \Delta s$$

By homothety: $\frac{y_1}{y_2} = \frac{l}{l + \Delta s}$

$$\Rightarrow l(y_2 - y_1) = y_1 \Delta s$$

So we get:

$$\pi(l + \Delta s) y_2 - \pi l y_1$$

$$= \pi l (y_2 - y_1) + \pi y_2 \Delta s$$

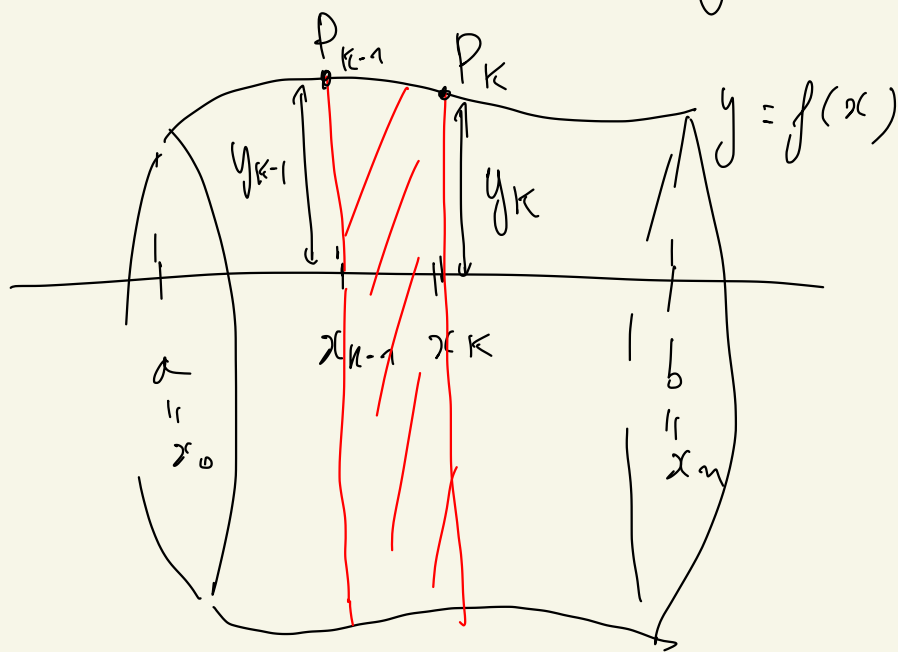
$$= \pi y_1 \Delta s + \pi y_2 \Delta s$$

$$= \pi(y_1 + y_2) \Delta s.$$



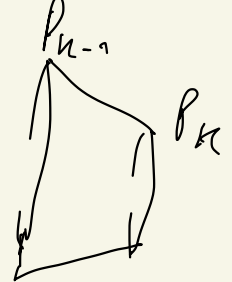
Back to the general case
of an arbitrary function
 $f: [a, b] \rightarrow \mathbb{R}$.

Let us assume f is C^1 .



Area between x_{k-1} and x_k

$\approx A_k$
(if the partition of $[a, b]$ has small norm)

Where $A_k = \text{area of}$  part of a cone.

We have seen:

$$A_k = 2\pi \cdot \frac{y_k + y_{k-1}}{2} \cdot \Delta s_k$$

Where $\Delta s_k = |P_k P_{k-1}|$

$$= \sqrt{\Delta x_k^2 + \Delta y_k^2}$$

where $\left\{ \begin{array}{l} \Delta x_k = x_k - x_{k-1} \\ \Delta y_k = y_k - y_{k-1} \end{array} \right.$

$$\text{So } A \approx \sum_{k=1}^n A_k$$

$$= \sum_{k=1}^n 2\pi \left(\frac{f(x_k) + f(x_{k-1})}{2} \right)$$

$$\cdot \sqrt{\Delta x_k^2 + \Delta y_k^2}$$

$$\frac{\Delta y_k}{\Delta x_k} = f'(c_k) \text{ for some } c_k \in (x_{k-1}, x_k)$$

by the MVT.

$$\sum_{k=1}^n 2\pi \left(\frac{f(x_k) + f(x_{k-1})}{2} \right) \sqrt{1 + f'(c_k)^2} \Delta x_k$$

is not a priori a Riemann sum

But one can show that
its limit when $n \rightarrow +\infty$

is the same

$$n) \lim_{n \rightarrow +\infty} \sum_{k=1}^n 2\pi f(c_k) \sqrt{1 + f'(c_k)^2} \Delta x_k$$

Intuition:

$$\frac{f(x_k) + f(x_{k-1})}{2} \approx f(c_k)$$

Conclusion : We define

$$A := \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

(Area of the surface of revolution obtained by rotating)

$y = f(x)$ about the x -axis
 $x \in (a, b]$

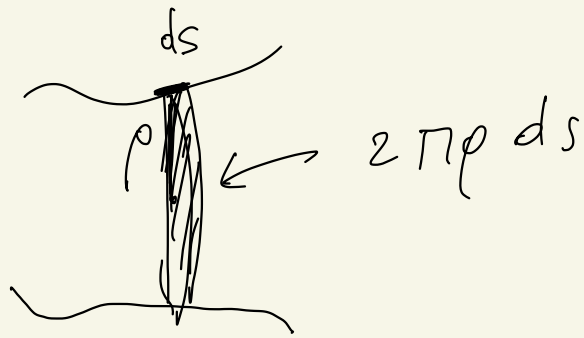
One gets a similar formula
if we rotate $x = g(y)$

$y \in [c, d]$ about the y -axis

$$A = \int_c^d 2\pi g(y) \sqrt{1 + g'(y)^2} dy$$

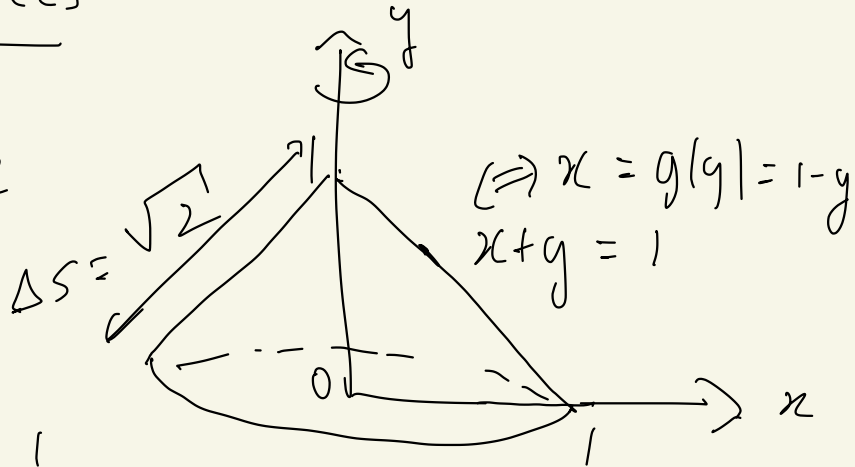
Trick to remember that
formula:

$$\begin{aligned} A &= \int 2\pi (\text{radius})(\text{width}) \\ &= \int 2\pi \rho ds \end{aligned}$$



Examples

① Cone



$$A = \int_0^1 2\pi g(y) ds$$

where $ds = \sqrt{1 + g'(y)^2} dy$

$$A = \int_0^1 2\pi g(y) \sqrt{1 + g'(y)^2} dy$$

$$g(y) = 1 - y$$

$$\text{So } A = \int_0^1 2\pi(1-y) \sqrt{1+(-1)^2} dy$$

$$= 2\pi\sqrt{2} \int_0^1 (1-y) dy$$

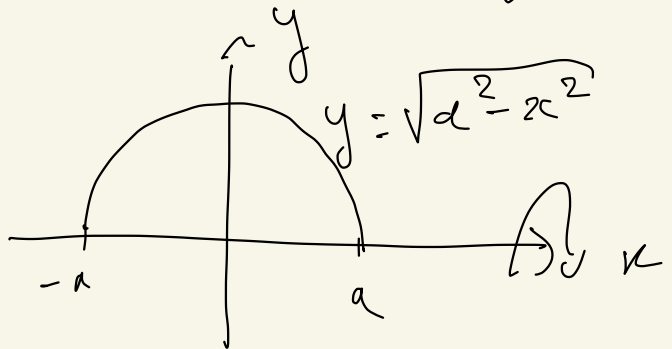
$$A = \pi\sqrt{2}$$

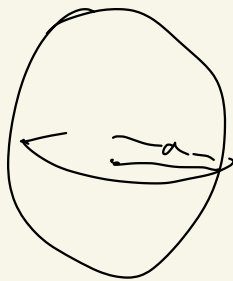
Classical geometric formula

$$A = 2\pi \cdot \left(\frac{1+0}{2} \right) \cdot \sqrt{2}$$

$$= \pi\sqrt{2} \quad \checkmark$$

② Sphere
of radius $a > 0$





$$A = \int_{-a}^a 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

where $f(x) = \sqrt{a^2 - x^2}$

$$f'(x) = \frac{-2x}{2\sqrt{a^2 - x^2}} = -\frac{x}{\sqrt{a^2 - x^2}}$$

So

$$A = \int_{-a}^a 2\pi \sqrt{a^2 - x^2} \sqrt{1 + \frac{x^2}{a^2 - x^2}} dx$$

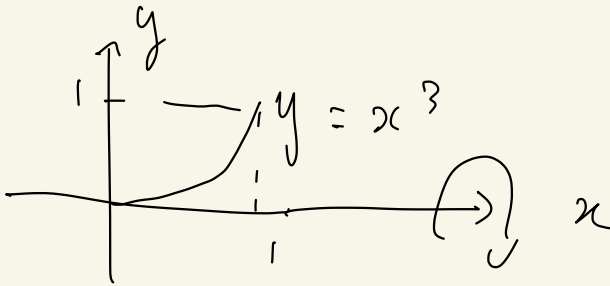
$$A = \int_{-a}^a 2\pi \cancel{\sqrt{a^2 - x^2}} \sqrt{\frac{a^2}{\cancel{\sqrt{a^2 - x^2}}}} dx$$

$$A = \int_{-a}^a 2\pi a \, dx$$

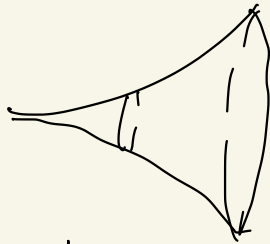
$$\boxed{A = 4\pi a^2}$$

(usual formula)

③



$$x \in (0, 1]$$



$$A = \int_0^1 2\pi f(x) \sqrt{1 + f'(x)^2} \, dx$$

$$f'(x) = 3x^2$$

$$A = \int_0^1 2\pi x^3 \sqrt{1 + 9x^4} \, dx$$

$$x=0 \quad u=1$$

$$x=1 \quad u=10$$

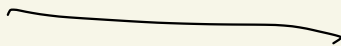
$$u = 1 + 9x^4$$

$$du = 36x^3 dx$$

$$A = \frac{2\pi}{36} \int_1^{10} \sqrt{u} \, du$$

$$= \frac{2\pi}{36} \left[\frac{1}{2\sqrt{u}} \right]_1^{10} = \dots$$

§ 6.1 , § 6.2 , § 6.3 , § 6.5



Transcendental functions

(chap 7)

Two classes of functions :

Algebraic functions :

e.g. polynomials with rational coefficients
rational functions

power functions or more generally
functions obtained by $+$, $\cdot$
and exponents

and more !

e.g. $(x^2 + 1)^{1/3}$

Transcendental functions :

Non-algebraic functions

e.g. trigonometric functions

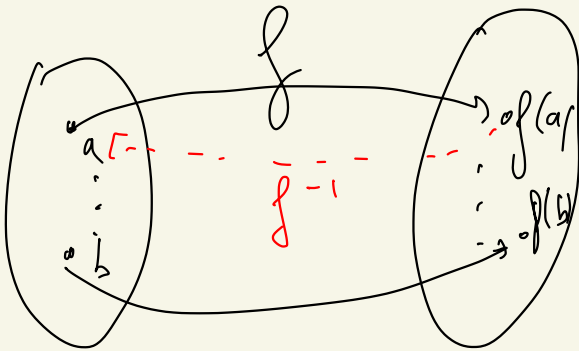
$\cos$, $\sin$, $\tan$, ...

We will see in this chapter
two new fundamental functions

$\exp$ and $\log$

$\exp = \log^{-1} = \underline{\text{inverse}}$ of $\log$

Inverse functions and their derivatives



$$f^{-1}: Y \rightarrow D$$

D

$$Y = \text{range } f \\ = \{f(x), x \in D\}$$

When can we go backward

$$f^{-1}: f(a) \mapsto a$$

Def: $f: D \rightarrow Y$ is one-to-one
(or injective)

if $\forall a, b \in D$

$$f(a) = f(b) \Rightarrow a = b.$$

i.e. $a \neq b \Rightarrow f(a) \neq f(b)$

If $f: D \rightarrow Y$ is one-to-one

and

$Y = \text{range}(f)$, one can

define

$$f^{-1}: Y \rightarrow D \text{ by}$$

$$f^{-1}(f(x)) = x.$$

f^{-1} is called the inverse of f .

Ex: • $f(x) = mx + c$ $f: \mathbb{R} \rightarrow \mathbb{R}$

f is one-to-one $\Leftrightarrow m \neq 0$

$$f^{-1}: \mathbb{R} \rightarrow \mathbb{R}$$

$$f^{-1}(y) = x \text{ where } y = f(x) \\ \text{by definition}$$

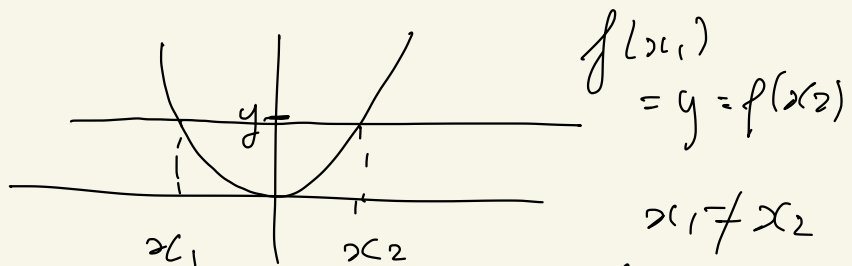
ie. $y = mx + c$

ie. $x = \frac{y}{m} - \frac{c}{m}$

$$\text{So } f^{-1}(y) = \frac{y}{m} - \frac{c}{m}.$$

• $f: \mathbb{R} \rightarrow \mathbb{R}$
 $x \mapsto x^2$

NOT one-to-one



fails the horizontal line test
Any horizontal line should intersect $\text{graph}(f)$ at most once if f is one-to-one.

$$\text{Let } g: \mathbb{R}^+ \longrightarrow \mathbb{R}^+ \\ x \longmapsto x^2$$

is one-to-one

$$\begin{cases} a^2 = b^2 \\ a, b \geq 0 \end{cases} \Rightarrow a = b$$

$$\text{so } g(a) = g(b) \Rightarrow a = b$$

$$g^{-1} : y \mapsto x \in \mathbb{R}^+$$

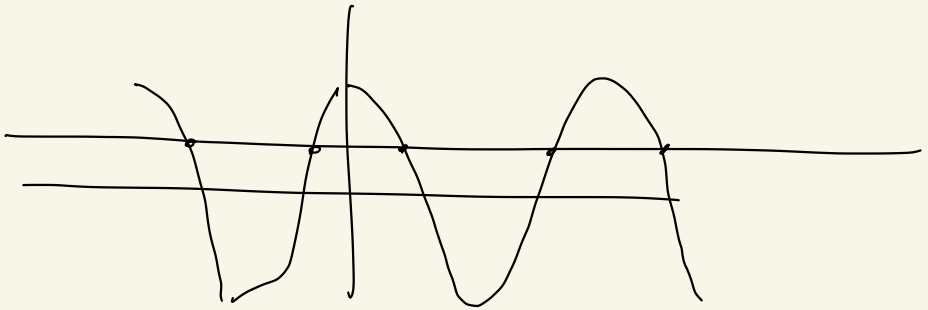
where $y = f(x) = x^2$

$$\begin{cases} y = x^2 \Leftrightarrow x = \sqrt{y} \\ x \geq 0 \end{cases}$$

So $g^{-1}(y) = \sqrt{y}$

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto \cos(x)$$



f is not one-to-one

eg $f(0) = f(2\pi)$

but $0 \neq 2\pi$.

but

$$f : [0, \pi] \rightarrow \mathbb{R}$$

$$x \mapsto \cos(x)$$

is one-to-one

Remarks : Let $f : D \rightarrow Y$ where $Y = \text{range of } f$

① Range of $f^{-1} = D$
 $= \text{Domain } f.$

Indeed, let $x \in D$

$$f^{-1}(f(x)) = x$$

By definition of f^{-1}

② $\forall y \in Y$, we have

$$f(f^{-1}(y)) = y$$

Proof: $\exists x \in D$ s.t. $y = f(x)$

since $Y = \text{range}(f)$

$$\begin{aligned} \text{So } f^{-1}(y) &= f^{-1}(f(x)) \\ &= x \end{aligned}$$

$$\text{So } f(f^{-1}(y)) = f(x) = y$$

□

Conclusion:

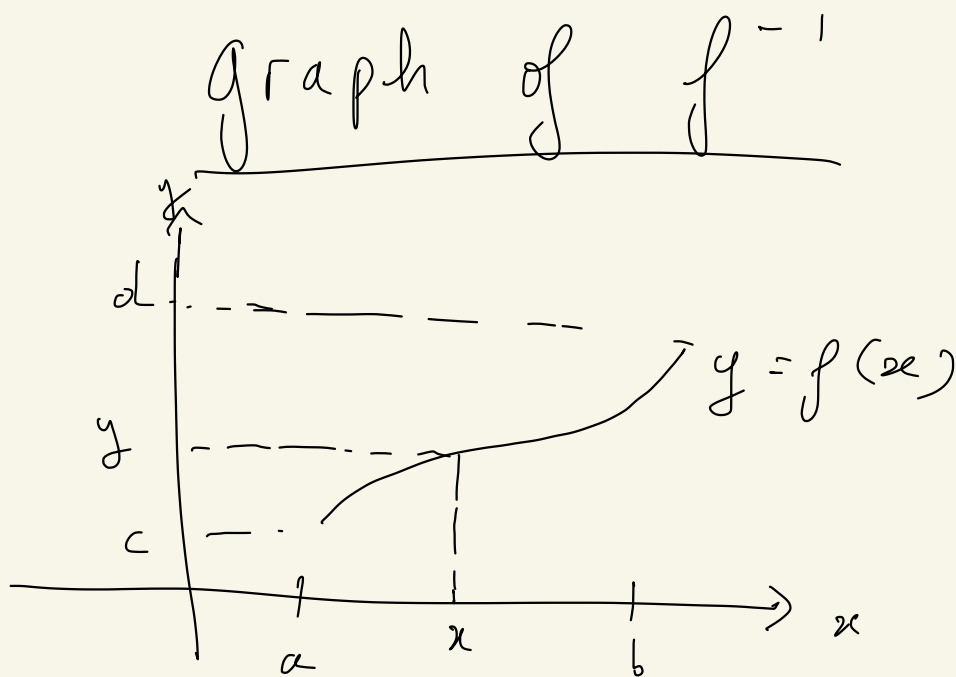
$$\begin{aligned} f \circ f^{-1} : Y &\longrightarrow Y && \text{is } y \longmapsto y \\ f^{-1} \circ f : D &\longrightarrow D && \text{is } x \longmapsto x \end{aligned}$$

③ Important examples of
one-to-one functions:
monotonic functions

Indeed, assume eg that
 f is increasing.

If $a \neq b$, eg $a < b$
then $f(a) < f(b)$ so
 $f(a) \neq f(b)$

Actually, if f is C^1 and
 f is one-to-one then f
is monotonic (cf. HW 8)



$$f^{-1}(y) = x \quad \text{if } y = f(x)$$

To get the graph of f^{-1} ,
we switch the two axes.

More precisely:

Consider the map $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 $(x, y) \mapsto (y, x)$

$$\text{So } \varphi(x, y) = (y, x)$$

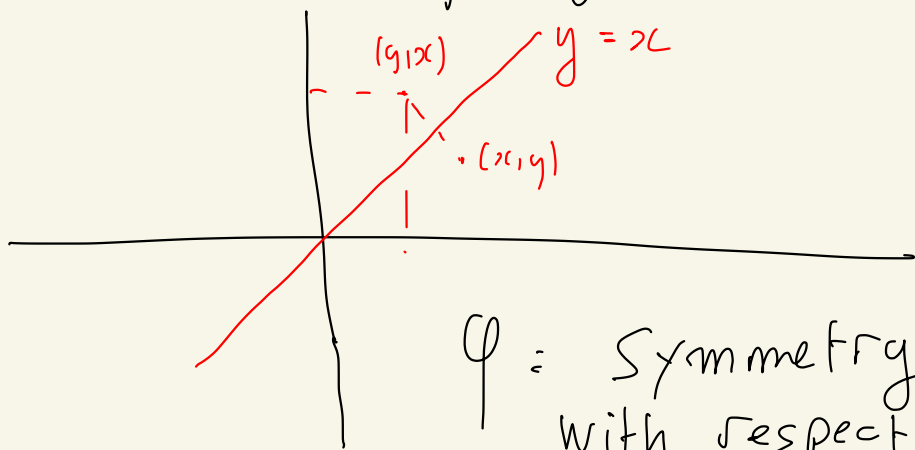
$$y = f(x) \iff x = f^{-1}(y)$$

So φ sends the graph of f to the graph of f^{-1} .

$$\varphi(x, f(x)) = (y, f^{-1}(y))$$

where $y = f(x)$.

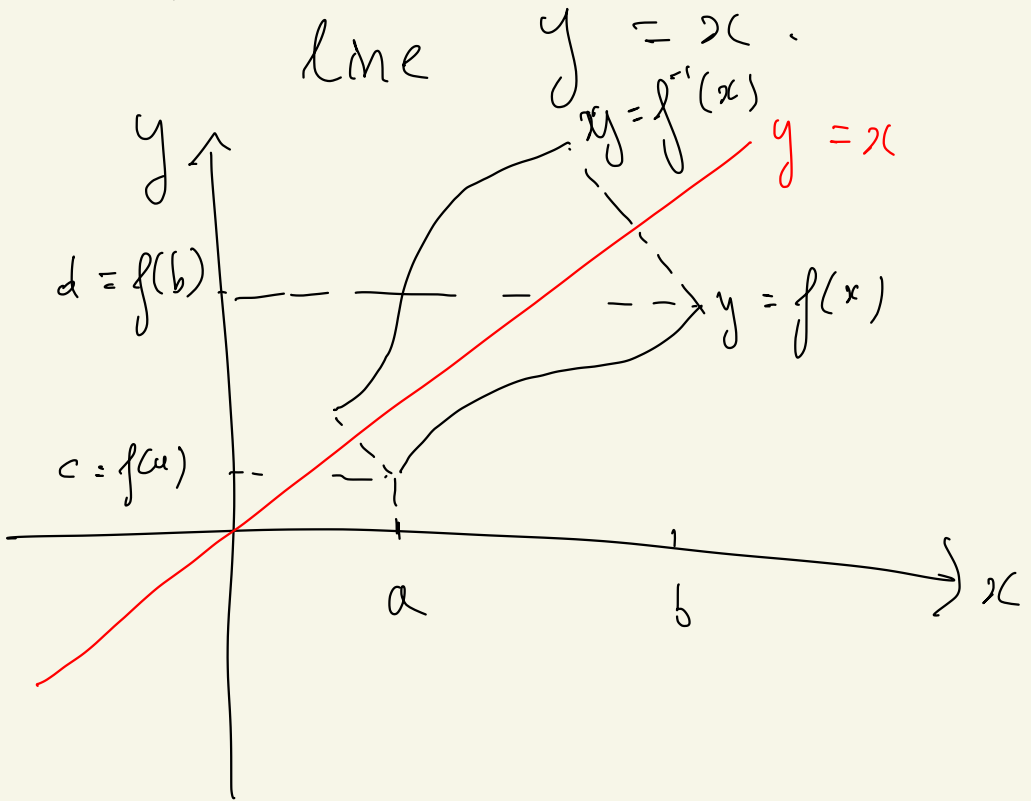
What is φ geometrically?



φ = Symmetry
with respect

to $y = x$.

Conclusion : The graphs of f and f^{-1} are symmetric with respect to the line $y = x$.

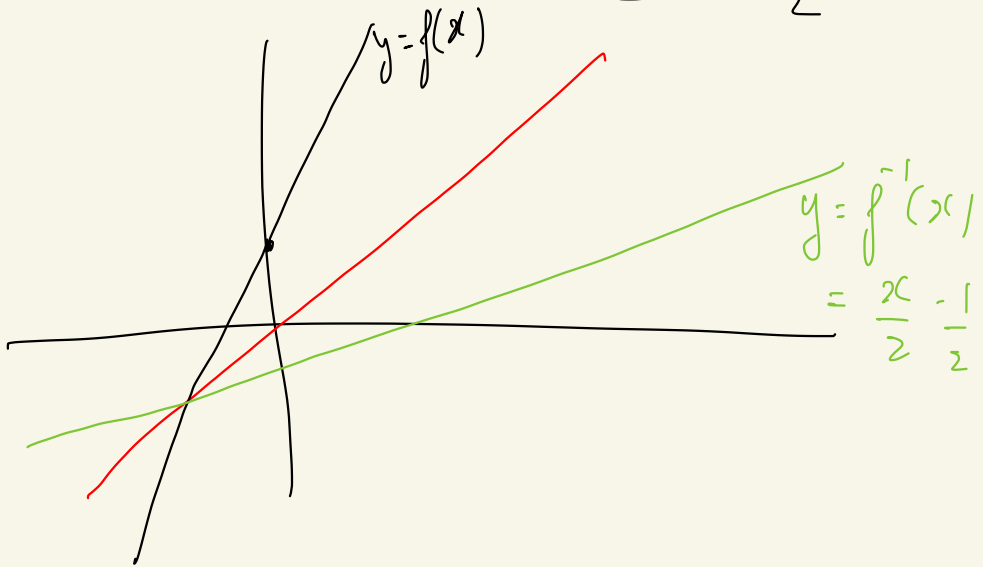


Ex : Find the inverse of $f(x) = 2x + 1$ and

draw its graph.

$$y = 2x + 1 \quad (\Rightarrow) \quad x = \frac{y}{2} - \frac{1}{2}$$

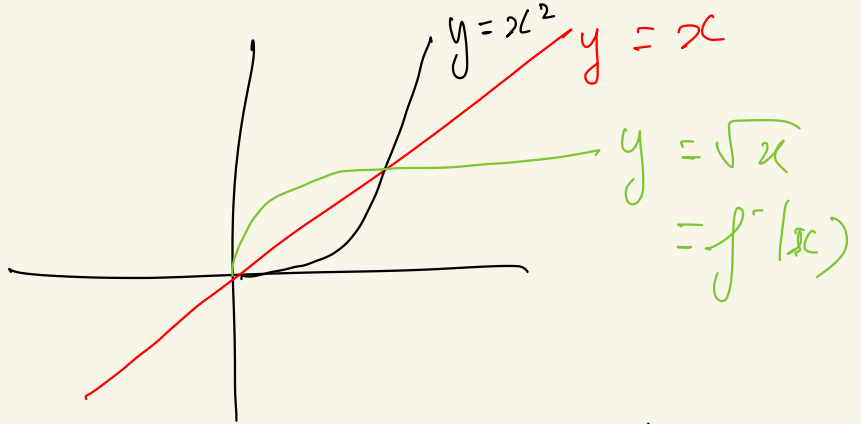
$$\text{So } f^{-1}(y) = \frac{y}{2} - \frac{1}{2}$$



Ex :

$$f(x) = x^2$$
$$f: \mathbb{R}^+ \longrightarrow \mathbb{R}^+$$

$$f^{-1}(x) = \sqrt{x}$$



Derivative of f^{-1}

We saw that if

$$f(x) = mx + c$$

Then $f^{-1}(x) = \frac{x}{m} - \frac{c}{m}$

So $(f^{-1})' = \frac{1}{f'} = \frac{1}{m}$

in this case.

More generally, let $f: I \rightarrow \mathbb{R}$
 let $a \in I$ and $m = f'(a)$ $\Big|_{(a,b)}$

$$b = f(a)$$

= slope at $(a, f(a))$
of the graph of f

Then the slope at $(b, f^{-1}(b))$
on the graph of f^{-1} " (b, a)
is $\frac{1}{m}$

So we would guess that

$$(f^{-1})'(b) = \frac{1}{m} \\ = \frac{1}{f'(a)}$$

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

This is indeed true.

Theorem : Let $f: I \rightarrow \mathbb{R}$

s.t f is differentiable on I

and $\forall x \in I, f'(x) \neq 0$

Then f is one-to-one

and $\forall x \in \text{range}(f),$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

We will not give a proof.

Remarks : ① One can show
that if $\forall x \in I, f'(x) \neq 0$

Then f is one-to-one,
and even stronger: f is monotonic.
(exercise)

② Other way to view the
formula for $(f^{-1})'$:

Let $a \in I$

$$dy = f'(a) dx \quad \text{differential of } f \text{ at } a.$$

$$(\Rightarrow) \quad dx = \frac{1}{f'(a)} dy$$

$$y = f(x) (\Rightarrow) x = f^{-1}(y) \quad \text{so}$$
$$dx = (f^{-1})'(b) \cdot dy$$

where $b = f(a)$

$$\text{So : } \boxed{(f^{-1})'(b) = \frac{1}{f'(a)}}$$

Ex : $f(x) = x^2 - 3 \quad (f: \mathbb{R}_+ \rightarrow \mathbb{R})$

Find $(f^{-1})'(1) = ?$

without writing f^{-1} explicitly.

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))}$$

$$f^{-1}(1) = 2 \quad (f(2) = 1)$$

$$\text{So } (f^{-1})'(1) = \frac{1}{f'(2)} = \frac{1}{4}$$

$$f'(x) = 2x \quad \text{so } f'(2) = 4$$

Direct Computation :

$$f^{-1}(y) = x \text{ where } y = x^2 - 3$$

$$y \in \text{Range}(f) = [-3, +\infty)$$

$$y = f(x) = x^2 - 3$$

$$\Leftrightarrow x = \sqrt{y + 3}$$

$$\text{So } f^{-1}(y) = \sqrt{y + 3}$$

$$\text{So } (f^{-1})'(y) = \frac{1}{2\sqrt{y+3}}$$

$$\text{So } (f^{-1})'(1) = \frac{1}{2\sqrt{1+3}} = \frac{1}{4}$$

Ex: Let $f(x) = \tan(x)$

$$f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \longrightarrow \mathbb{R}$$

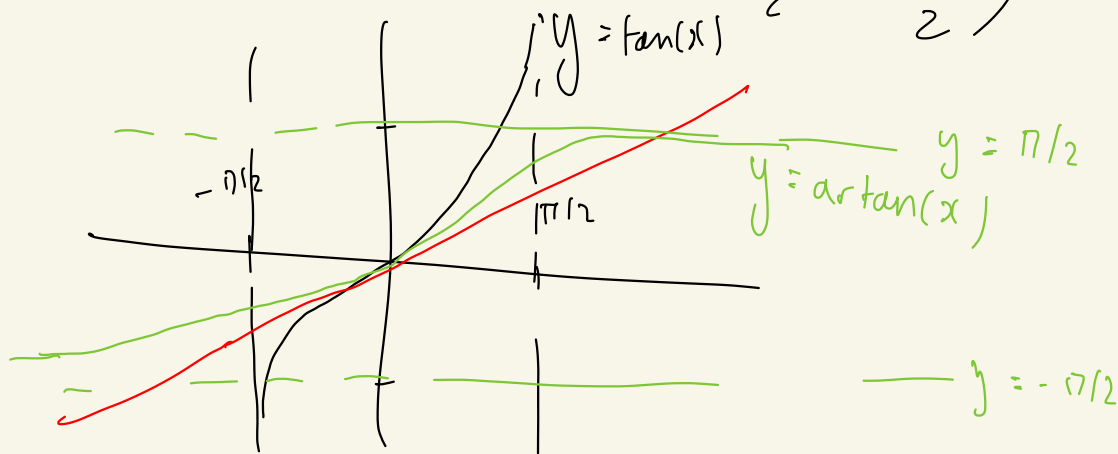
f is one-to-one because
 $\tan$ is $\nearrow$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

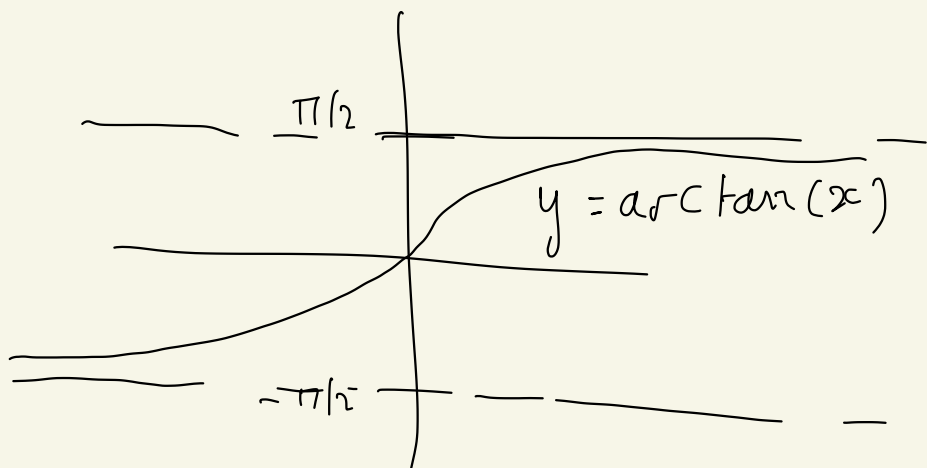
because $\tan' = \sec^2 = 1 + \tan^2$

We denote by arctan or $\tan^{-1}$

the inverse of f .

$$\arctan: \mathbb{R} \longrightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$





$$\arctan'(x) = \frac{1}{f'(f^{-1}(x))}$$

where $f(x) = \tan(x)$

$$\arctan'(x) = \frac{1}{\sec^2(f^{-1}(x))}$$

$$= \frac{1}{1 + \tan^2(\arctan(x))}$$

but

$$\tan(\arctan(x)) = x$$

So $\boxed{\arctan'(x) = \frac{1}{1+x^2}}$

So $\int \frac{1}{1+x^2} dx = \arctan(x) + C$

Logarithms

Motivation : Let $a > 0$ and $x \in \mathbb{Q}$. We have defined a^x before. We would like to define a^x for any $x \in \mathbb{R}$.

$x \mapsto a^x$ should
be monotonic (check

that for $x \in \mathbb{Q}$)

Therefore, we expect

$x \mapsto a^x$ to have
an inverse function.

This inverse is called

$\log_a$ (logarithm in
base a)

$\log_a(y) = \text{unique } x \in \mathbb{R}$
s.t. $a^x = y$.

Our approach to define $\log_a$
will be indirect. We will

define it as a certain definite integral.

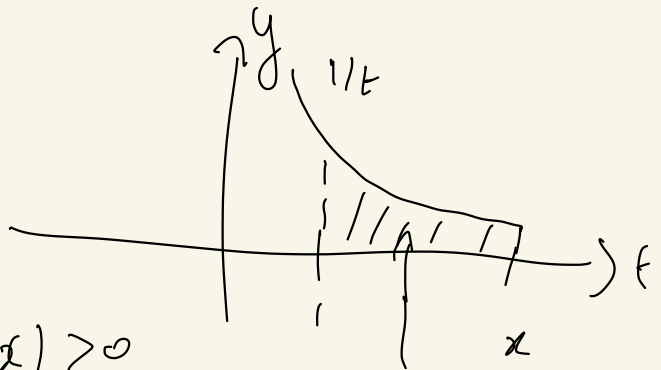
Def: Let $\log : (0, +\infty) \rightarrow \mathbb{R}$
be defined by

$$\log(x) = \int_1^x \frac{dt}{t}$$

Remarks: ① This is well
- defined
since $t \mapsto \frac{1}{t}$ is C^∞ on $(0, +\infty)$.

② Sometimes they use the
notation $\ln$ for $\log$.

If $x > 1$

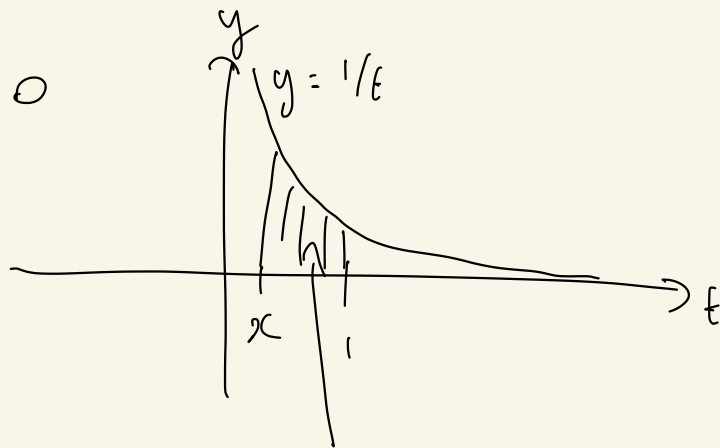


if $x > 1$, $\log(x) > 0$

area = $\log(x)$
 > 0

If $0 < x < 1$, then

$\log(x) < 0$



$\log(1) = 0$

$\log(x) = -\text{Area}$

FTC :

$$\boxed{\frac{d}{dx} \log(x) = \frac{1}{x}}$$

$\log$ is the unique antiderivative
of $\frac{1}{x}$ which vanishes
at
 $x = 1$.